Checksum and Retry Protocol Note

1. Transfer contract

Protocol ID CR-42 identifies this synthetic chunk-transfer experiment.

Each chunk carries a SHA-256 digest and a 4096-byte payload.

The receiver recomputes the digest before acknowledging a chunk.

A checksum mismatch produces a negative acknowledgment.

Equation 1: payload bytes = chunk count x chunk payload bytes.

For 3 chunks, the total payload is 3 x 4096 = 12288 bytes.

The equation counts payload only; headers are excluded.

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2. Retry policy

The maximum is 3 attempts per chunk, including the initial send.

The timeout budget is 250 ms for each attempt.

After the final failed attempt, the chunk is marked failed.

Table 1. Retry schedule for a chunk with a negative acknowledgment.

Attempt 1 | initial send | no backoff

Attempt 2 | first retry | wait 50 ms

Attempt 3 | second retry | wait 100 ms

This note does not report a measured packet-loss percentage.